Multi-Agent AI Workflow Engine: Simulating an Autonomous Software Development Team

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Abstract

Most AI coding assistants work as a single model answering a single prompt: one pass to understand the request, design a solution, write the code, and check its own work. This project asks whether dividing that work among specialized agents produces more correct software, and at what cost.

We built MAWE, a Multi-Agent AI Workflow Engine that splits development across three agents. A Product Manager turns a plain-language request into a specification with acceptance criteria and runnable checks. A Developer writes the code. A Code Reviewer runs that code in a sandbox and compares its output against the specification before approving it or sending it back. Because the Reviewer executes the program instead of reading it, a model cannot approve code that demonstrably fails.

Two problems follow. Because two agents can disagree indefinitely, the loop has five independent exit conditions; across 100,000 randomized adversarial sequences, every run terminated within its iteration limit. Because the conversation grows with every revision, older iterations are compressed into a structured log, cutting prompt size by 94 percent at 80 iterations without changing the outcome.

For comparison against simpler approaches, we built a benchmark of 20 Python and JavaScript tasks with 181 hidden tests the agents never see, so scoring does not depend on the reviewing model. The benchmark was validated rather than assumed: every task is solvable, none can be passed by a program that does nothing, and introduced defects are caught 91 percent of the time.

Whether role separation improves correctness, and whether any improvement justifies its cost, requires a comparison against a live language model that has not yet been run.